



# Junaid Ul Ahsan

PhD, Physics,  
NIT Srinagar

- October-19-1994
- NIT, Srinagar, 190006 J & K, India
- +918491023854/+919906719901
- ahsanjunaid57@gmail.com
- INDIAN

## Social Network

- Researchgate Link
- Google Scholar Link

## Languages

- Hindi/Urdu
- English

## SOFTWARES

- Vampire
- Quantum Espresso
- Wien2K
- VASP

## Memberships

Asian Consortium on Computational  
Materials Science (ACCMS)

“My current research lies at the intersection of **computational materials science** and **atomistic modeling**, with a strong emphasis on low-dimensional and nanostructured materials. I investigate material behavior at the atomic scale by employing **density functional theory (DFT)** to compute ground-state energies, optimize crystal structures, and analyze electronic band structures and densities of states, with the broader aim of understanding structure–property relationships relevant to functional materials.”

## Teaching Experience

- 2024–2025 **Quantum Mechanics, Classical Mechanics** (Govt. Degree College Tangdar, Jammu & Kashmir India)
- 2021–2023 **MATLAB, Classical Mechanics** (M.Sc — NIT Srinagar)
- 2019–2022 **General Lab B.Tech** — NIT Srinagar
- 2017–2019 **General Lab, Tutorial** (B.Sc & M.Sc — Panjab University Chandigarh)

## Education

### Postgraduate Training

- 2019 – 2024 **PhD, Physics** **National Institute of Technology, Srinagar**  
Thesis title: Magnetic properties of transition metal-based nanostructures using atomistic simulations
- 2017 – 2019 **Junior Research Fellow** **Panjab University, Chandigarh**  
Nuclear Structure study

### Study

- 2014 – 2016 **Master of Science** **AMU Aligarh**  
Department of Physics
- 2011 – 2014 **Bachelor of Science (Honours)** **AMU Aligarh**  
Department of Physics

## Awards

- 2026 Jammu Kashmir State Eligibility Test
- 2014 National Graduate Physics Examination
- 2016 Graduate Aptitude Test in Engineering

## Scholarships

### TEEP

- 2023 **Taiwan Education Experience Program** **National Taiwan University**  
December 2022 — May 2023

## Technical Expertise

### Scientific and Computational Tools

- Atomistic Spin Dynamics** **Vampire Software**  
Magnetic nanomaterials simulation
- Density Functional Theory** **Quantum Espresso, Wien2K, VASP**  
Electronic structure calculations
- Visualization & Data Analysis** **Origin, Gnuplot, Vesta, xmgrace**  
Scientific plotting and crystal visualization
- Programming** **Python, MATLAB**  
Numerical modelling and data analysis
- Workflow Platforms** **AIIDA, Boltztrap, Gibbs**  
High-throughput calculations and thermoelectric modelling

### Operating Systems and Documentation

- Operating Systems** **Linux, Windows**  
Scientific computing and scripting environments
- Scientific Writing** **LaTeX, MS Word, PowerPoint, Excel**  
Research documentation and presentation

**Scientific Communication**

Paper Writing, Reviewing & Editing

Journal publications and peer review

**Research Skills**

Independent research, Collaborative research

Project planning and interdisciplinary collaboration

## Publications

### Journal Articles

- 2024 **Exploring the supercapacitive potential of Sm<sup>3+</sup> doped CoFe<sub>2</sub>O<sub>4</sub> nanoparticles: Enhanced magnetic and dielectric properties**  
*Bhat, S. A., Tantray, A. M., Ahsan, J. U., & Ikram, M.*  
Ceramics International, 50(14), 26220-26233.
- 2022 **Temperature dependent magnetization in Co@Fe Nanoparticles**  
*Ahsan, J. U., & Singh, H.*  
Physica B: Condensed Matter, 627, 413488.
- 2022 **Atomistic simulation study of FeCo alloy nanoparticles**  
*Ahsan, J. U., & Singh, H.*  
Applied Physics A, 128(5).
- 2023 **Magnetic features of hybrid transition metal-rare earth nanoparticles: Monte Carlo simulations**  
*Ahsan, J. U., & Singh, H.*  
Physica B: Condensed Matter, 668, 415223.
- 2024 **Magnetic-ordering temperature of cobalt/nickel bimetallic nanoparticles: a Monte Carlo investigation**  
*Ahsan, J. U.*  
Applied Physics A, 130(2), 124.
- 2023 **Atomistic study of Positive Exchange Bias in Hard/Soft ferromagnetic bilayer nanosystem**  
*Ahsan, J. U., & Singh, H.*  
Physica Scripta, 98(2), 025403.
- 2021 **Random oriented ZnO nanorods fabricated through anodization of zinc in KHCO<sub>3</sub> electrolyte**  
*Tantray, A. M., Mir, J. F., Mir, M. A., Rather, J., & Shah, M. A.*  
ECS Journal of Solid State Science and Technology, 10(8), 081003.
- 2025 **In-depth study of double perovskite Sr<sub>2</sub>NiTaO<sub>6</sub>: Structural, electronic, thermoelectric, and spintronic properties for sustainable and high-performance applications**  
*Ahsan, J. U., Rather, M. R., Sultan, K., Zriouel, S., & Hill, E.*  
Computational Condensed Matter, 43, e01026.
- 2023 **Atomistic simulation of Stoner–Wohlfarth (SW) particle**  
*Ahsan, J. U., Majeed, A., & Singh, H.*  
International Journal of Materials Research, 114(7-8), 641-645.
- 2022 **Manipulating Superconductivity in Superconductor/Ferromagnet Hybrid Nanostructures**  
*Majeed, A., Ahsan, J. U., & Singh, H.*  
International Conference on Nanotechnology: Opportunities and Challenges.
- 2025 **Exchange bias study in FM/AFM bilayers: Atomistic simulations**  
*Ahsan, J. U.*  
Physics Open, 23, 100265.
- 2019 **Shell-model description in <sup>99</sup>Rh and systematics of odd-A Rh isotopes**  
*Kumar, S., Sihotra, S., Singh, V., Rather, J., Kaur, M., Goswamy, J., & Singh, N.*  
Acta Physica Polonica B, 50(2), 159.

- 2017 **Band Structures in  $^{100}\text{Pd}$**   
*Kaur, M., Sihotra, S., Singh, V., Rather, J., Singh, N., Biswas, S., & Palit, R.*  
Proceedings of the DAE Symposium on Nuclear Physics, 62, 92.
- 2018 **Band Structure in  $^{120}\text{I}$**   
*Kaur, M., Sihotra, S., Rather, J., Singh, V., Kumar, S., Singh, N., & Kaur, N.*  
Proceedings of the DAE International Symposium on Nuclear Physics, 63, 120.
- 2017 **Rotational Band Structure in  $^{101}\text{Pd}$**   
*Rather, J., Bhat, G. H., Singh, V., Sihotra, S., Mehta, D., & Sheikh, J. A.*  
Proceedings of the DAE Symposium on Nuclear Physics, 62, 226.

## Workshops & Conferences

- 2026 **UGC–Malaviya Mission Teacher Training Centre (MMTTC)**  
*Short Term Programme on Role of Artificial Intelligence and Machine Learning in Material Sciences*  
**Hands-on Training Programme**
- 2021 **World Nano Congress on Advanced Science and Technology (WNCST)**  
*Centre for Nanotechnology Research (CNR), VIT Vellore*  
**Poster Presentation**
- 2021 **International Conference on Nanoscience and Nanotechnology (ICONN 2021)**  
*SRM Institute of Science and Technology (Virtual)*  
**Paper Presentation: Temperature dependent magnetization in FeCo alloy nanoparticles**
- 2021 **International Conference on Nanoscience and Nanotechnology (ICONN 2021)**  
*SRM Institute of Science and Technology (Virtual)*  
**Paper Presentation: Study of Exchange Bias in Co/CoO nanobilayer system**
- 2021 **20th International Workshop on Computational Physics and Materials Science: Total Energy and Force Methods**  
*ICTP, Italy (Virtual)*
- 2021 **Magnetism 2021**  
*UK and ROI Magnetism Community (Virtual)*
- 2021 **ICEAM 2021**  
*Jaypee Institute of Information Technology, Noida (Virtual)*
- 2020 **Short Term Course on Current Trends in Condensed Matter Physics**  
*Dr. B. R. Ambedkar National Institute of Technology, Jalandhar*  
(TEQIP-III Sponsored, 25–29 Sept.)
- 2020 **Winter School on Material Characterization Techniques**  
*Manipal University Jaipur*  
(Under International Conference on Multifunctional Nanomaterials – ICMN)
- 2017 **International Conference in Nuclear Physics with Energetic Heavy Ion Beams**  
*Panjab University, Chandigarh*
- 2017 **DAE Symposium on Nuclear Physics**  
*Thapar University, Patiala*

## References

- Dr. Harkirat Singh **Ph.D. Supervisor**  
National Institute of Technology, Srinagar  
Email: harkirat@nitsri.ac.in
- Prof. Devendra Mehta **Project Supervisor**  
Panjab University Chandigarh  
Email: dmehta@pu.ac.in
- Prof. Arti Kashyap **PhD Viva Voce Examiner**  
Indian Institute of Technology Mandi  
Email: arti@iitmandi.ac.in